NATHAN BERNIER

STRENGTHS

- Algorithms and signal processing for high-precision sensors
- Mathematical modeling and numerical simulations
- 10+ years of development with Python
- Statistical analysis with Bayesian inference
- Quick to learn and master new subject

EDUCATION

2019 PhD in Experimental Physics
EPFL (Swiss Federal Institute of Technology, Lausanne)

2013 MSc Physics
EPFL (Swiss Federal Institute of Technology, Lausanne)

- Master thesis at **Harvard University** published as article
- Best average in graduating BSc class (71 students)
- Exchange year at **Carnegie Mellon University**

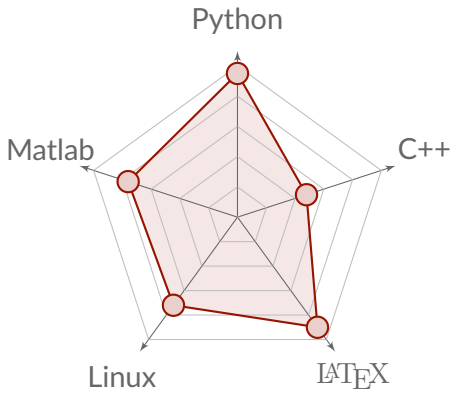
CONTACT

nathan-rafael-bernier
nrbernier.github.io
nathan@bernier.ch
+41 79 580 87 49

LANGUAGES

French: C2 (native speaker)
English: C2 (5+ years of work in English)
German: B1

PROGRAMMING



ACHIEVEMENTS

- Patent co-author on miniature 3D camera (pending)
- 8 publications in scientific journals including Nature Physics (full list: ORCID 0000-0002-5837-8407)
- EPFL Excellence Fellowship 2011 & 2012

EXPERIENCE

2025 – 2023

Research Engineer
Atracsys (Smith & Nephew)
2.5 years
Puidoux, Suisse

R&D metrology for 3D cameras with medical applications

- Autonomous project on novel miniature camera:
  - Image-analysis algorithms and optical models (Python/Matlab) to optimize performance for surgical applications
  - Self-made prototype to test image sensor, with optical mask alignment down to tenth of degree
  - Optical simulations leading to the discovery of optical aberrations from the image sensor at wide incidence angles
- Full noise simulation of cameras (Python) leading to new proposed optical markers with 20% performance increase
- Measurement and calibration of cameras using a CMM
- Client support on metrological questions

EXPERIENCE

2023

Senior Scientist
Miraex

2.5 years

Ecublens, Suisse

R&D for vibration and pressure photonic sensors

- Development of embedded algorithms optimized for real time (Python): noise cancellation and frequency filtering
- Conception and implementation of new calibration setup with measurement automation (Python)
- Physical models of noise sources to understand limits of performance and help dimension electronics
- Technical support on site for pilot projects: wind tunnel, factory machines and train transportation
- IP: competition watch and liaison with patent attorney

2020

PhD researcher
EPFL

4.5 years

Lausanne, Suisse

- Managing a research project in collaboration with a team of 4, leading to 7 scientific publications over 5 years
- Lead on data analysis and visualization (Python)
- Mathematical models of the circuits with original theoretical predictions confirmed by experiment
- Design and simulation of microwave circuits (Sonnet) to optimize their properties for the research goals
- Precision electronic and optical measurement with instrument automation (Python)
- Handling of cryostat and samples (cryogenics, wire bonding)
- Posters and oral presentations at conferences in Switzerland, France, Italy, Canada and in the USA

2014

SKILLS

Transversal

- Clear communication adapted to audiences of varied backgrounds
- Autonomous project management
- Easy collaboration in multidisciplinary teams

Technical

- Mathematical models and simulations for electronics, optics and noise properties
- Algorithms for signal processing (DSP) for series-data and images
- Precision laboratory measurements for electronics and optics
- Data analysis and visualisation

ADDITIONAL EXPERIENCE

Postdoc
EPFL

2019: 6 months

Lausanne, Suisse

- Managing a research project for a team of 3
- Development of a Python module for function fitting with Bayesian inference

ACTIVITIES

- Photography
- Cycling and running
- Board games
- Sewing and woodwork

PERSONAL DETAILS

Lausanne, Switzerland
Born 1990
Citizenship: Swiss, French, Canadian